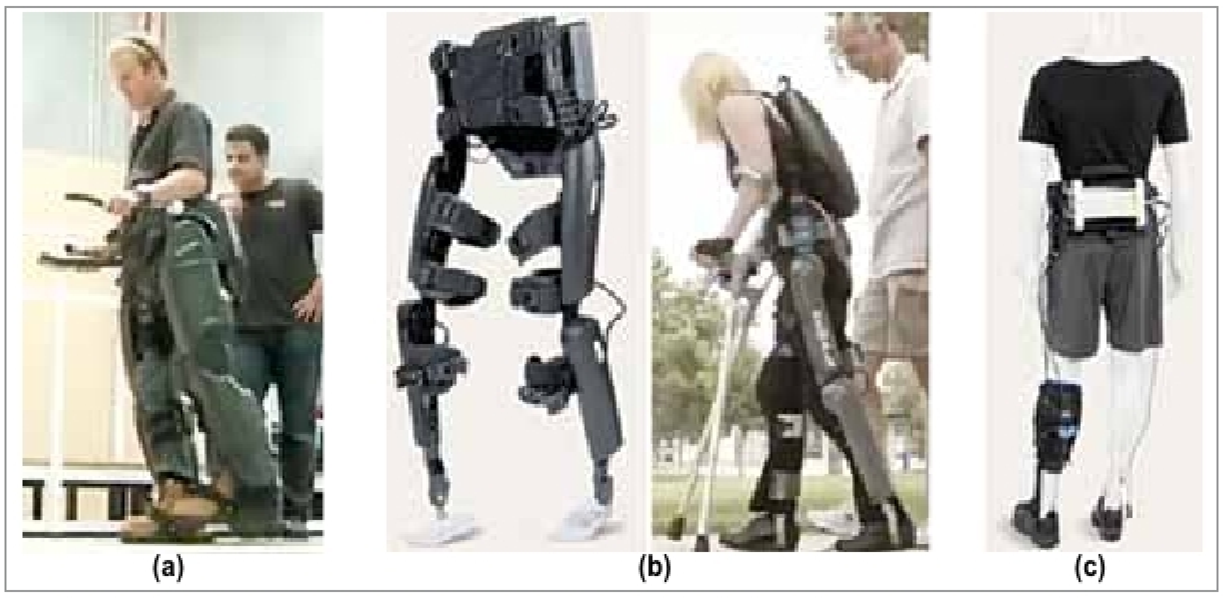

Fig. 8: (a) Fully automated exoskeleton for paraplegics (Credit: Rex Bionics), (b) exoskeleton for spinal cord injury, (c) exoskeleton for stroke (Credit: ReWalk)

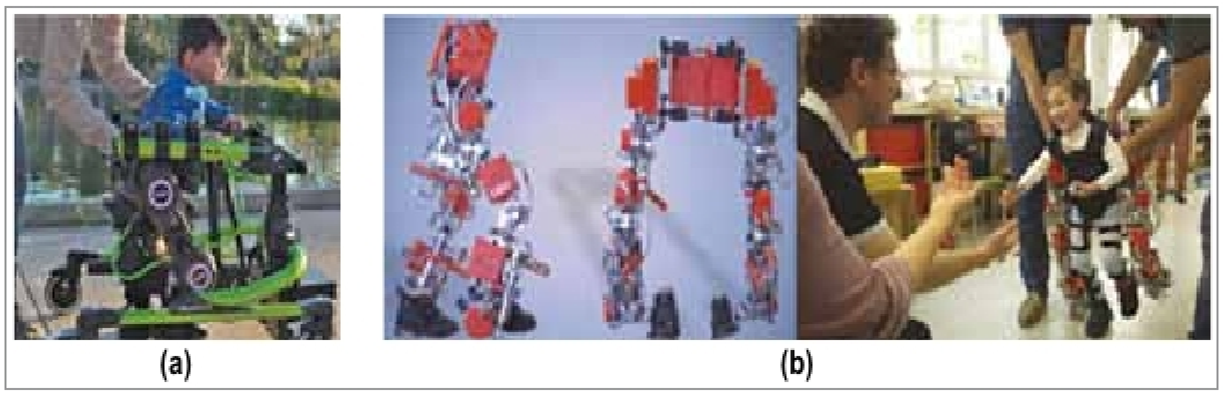

Fig. 9: (a) Exoskeleton for cerebral palsy (Credit: trexo), (b) exoskeleton for spinal muscular atrophy in children (Credit: Marsi Bionics CSIC)

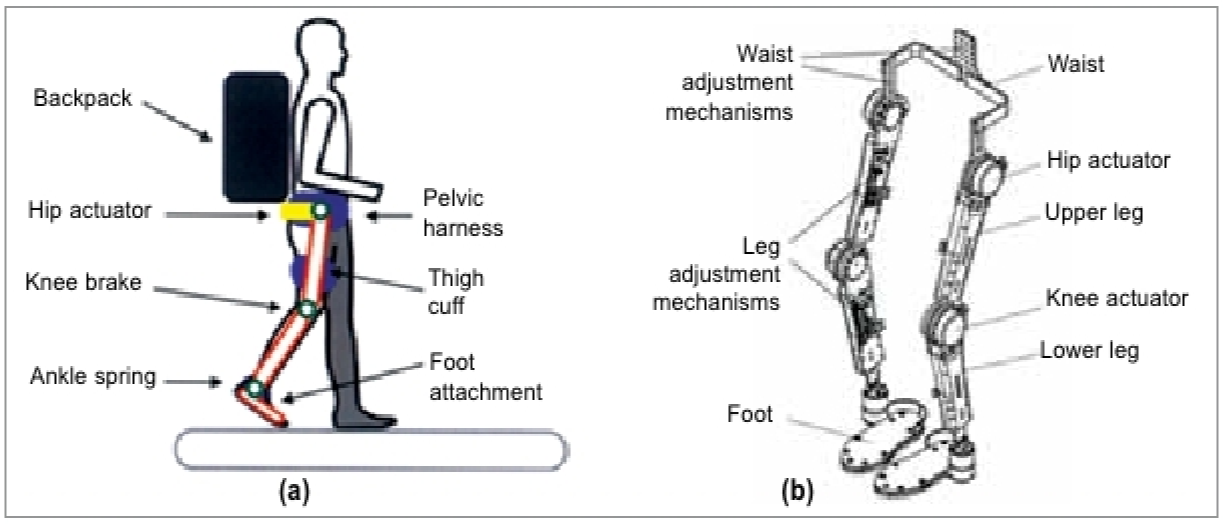

Fig. 10: (a) Main components of the exoskeleton (Credit: James Conor Walsh), (b) design of lower-limb exoskeleton (Credit: intechopen.com)

and sensors. The mechanical design of each exoskeleton varies depending on the patient's level of support. Based on ergonomics, the material used is usually metal, which has to be light and flexible, thus making it easy and natural to wear.

Some exoskeletons are also made of soft materials that can be wrapped around the affected parts like wrist, upper arm, etc. Some exoskeletons require electrical power, and some can be used without any power, giving more degrees of freedom to the wearer. Also, the frames are powered in different ways, that is, mechanically, electrically, or using batteries.

Different companies use different robotic gadgets and different adjustable critical parts as per their customers' needs. The actuators used should also be compact and strong. In addition, these should be capable of meeting the speed and movement requirements for the targeted motion, such as sitting, standing, and walking, to minimise any inconvenience to the wearer.

Actuators can be electric, pneumatic, hydraulic, shaped memory alloy (SMA), or passive. Selecting the joints should be done carefully to achieve a range of degrees of freedom for a range of motion as per the human anatomy.

In case of the upper limb, exoskeleton joints are needed for shoulders, elbow, and wrist. In case of the lower limb, exoskeleton joints are needed for the hip, knee, and ankle. Leading companies increase the range of degrees of freedom of joints and the range of motion so that the user feels comfortable.

However, for patients who have lost their muscular activity in their limbs, the range of degrees of freedom should be restricted to prevent uncontrolled movements. Manufacturers will perform a Clinical Gait Analysis (CGA) to determine the required degrees of freedom and also calculate the required joint movements for the targeted motion. If joints are to be powered, it is done by DC motors.

Control

Controlling the exoskeleton is based on the human-exoskeleton interaction, which is done using interaction signals between the user and the device. But, unfortunately, many paraplegic patients are not able to generate a human-exoskeleton interaction. In such cases, some companies use electromyography (EMG) in determining the user's intentions. But EMG produces noise and also requires extensive signal conditioning.

Pre-defined motion control (PMC) is more suitable because the user can select the intended motion, thereby reducing the computational and hardware complexity.

Feedback

The exoskeleton has a number of input signals based on two features-structural feedback and human feedback. Structural feedback is associated with actual movements and relative position of the mobile parts of the exoskeleton, and human feedback is associated with the biological electrical signals generated by the human nervous system due to the action of exoskeleton on the human body and user's intention to move.

Signals based on structural feedback are obtained from rotation encoders, force sensors torque sen-